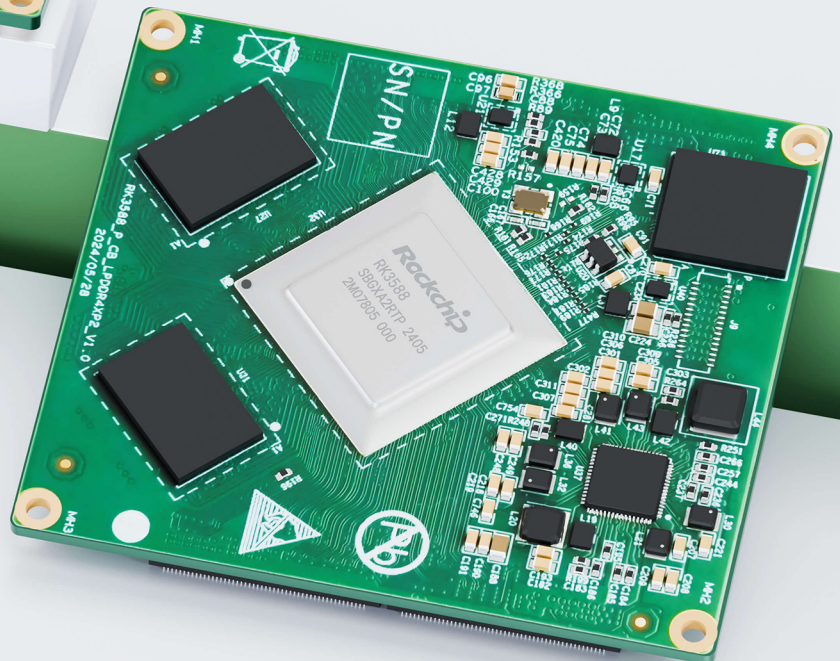
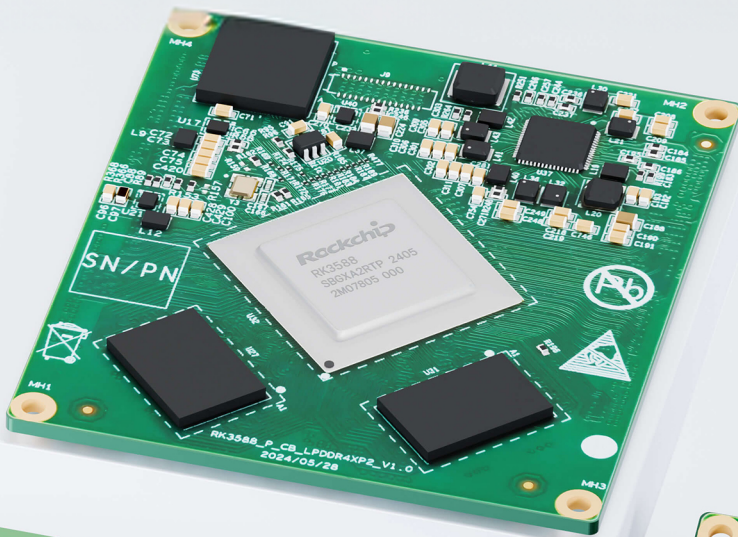


Embrace Edge AI, Empower Industrial Digitalization



EC3588-C AI System on Module

· RK3588 ARM SoC

· 4×A76+4×A55

· 6TOPS NPU

· 8K Video Codec

1. Product Overview

EC3588-C system on module is developed based on the Rockchip RK3588 flagship ARM SoC, delivering powerful performance for high-end embedded applications.

Product Features:

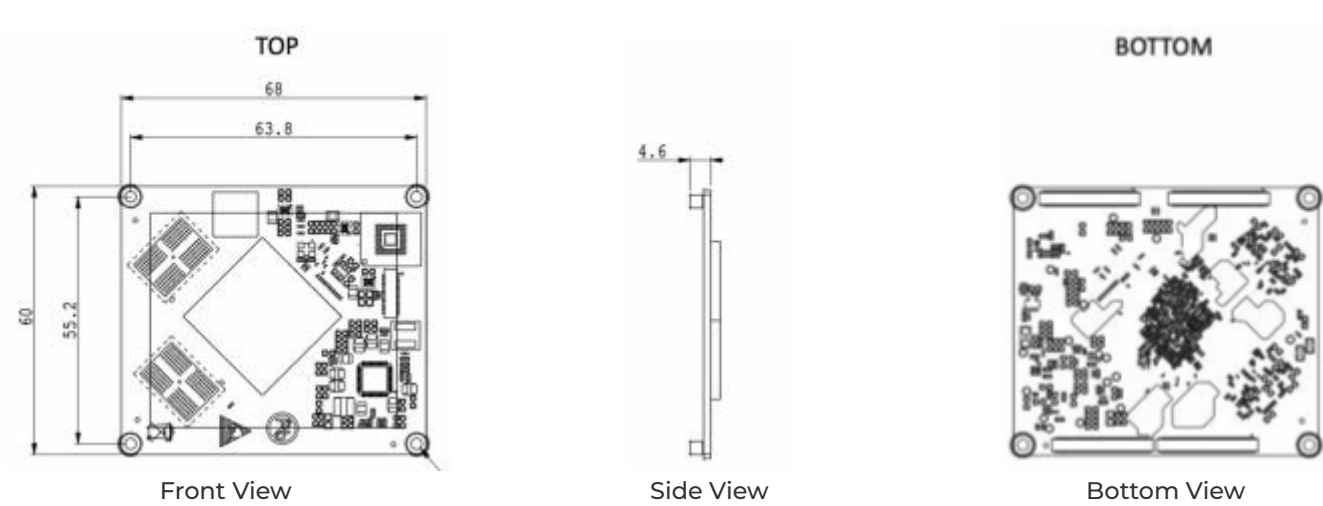
- **High-Performance CPU:** 8nm octa-core, 4×A76+4×A55 @ 2.4 GHz
- **Powerful AI NPU:** 6.0 TOPS triple-core, multi-format acceleration
- **Advanced Graphics:** Mali-G610 MP4, 8K@60fps codec, quad-screen output
- **Rich Interfaces:** PCIe 3.0, USB 3.1, SATA, GMAC, MIPI-CSI/DSI
- **High-End Media:** HDMI RX/TX, DP TX, multi-camera support

Core Technical Specification

Technical Indicator	Specification
OS	Linux
Industrial Protocol	Modbus RTU/TCP, EtherNet/IP, OPC UA, Mitsubishi MC
Remote Management	InHand DeviceLive, HTTPS, SSH
AI	6.0 TOPS NPU
Video Codec	8K@60 fps decode, 8K@30 fps encode
Cloud Platform	Cloud parameter config, container management, firmware management
CPU	4×A76 @ 2.4 GHz + 4×A55 @ 1.8 GHz
Memory / Storage	16 GB LPDDR4 / 64 GB eMMC
Display Interface	HDMI/eDP TX, DP TX, MIPI DSI, quad independent display

Technical Indicator	Specification
Connectivity	2×GMAC, USB 3.1, PCIe 3.0, SATA, MIPI-CSI
Dimensions (W × D × H)	68 × 60 × 4.6 mm (2.68 × 2.36 × 0.18 in)
Operating Temperature	0 °C ~ +80 °C (32 °F ~ +176 °F)

2. Product Dimensions



- Note:
1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
 2. All dimensions are approximate and for reference only.
 3. Dimensioned drawings are not intended for machining.
 4. Dimensions are subject to part and manufacturing tolerances.
 5. Specifications may change without prior notice.

3. Hardware Specifications

Category/Parameter	Specification
Hardware Platform	

Category/Parameter	Specification
CPU	4 × ARM Cortex-A76 @ 2.4 GHz + 4 × ARM Cortex-A55 @ 1.8 GHz
NPU	6.0 TOPS, triple core, supports int4/int8/int16/FP16/BF16/TF32 acceleration
GPU	ARM Mali-G610 MP4, OpenGL ES 1.1/2.0/3.2, OpenCL 2.2, Vulkan 1.2
VPU	Decode: H.265/VP9 up to 8K@60 fps, H.264 up to 8K@30 fps, AV1 up to 4K@60 fps; Encode: H.265/HEVC, H.264/AVC up to 8K@30 fps
RAM	16 GB LPDDR4
ROM	64 GB eMMC
Video Input	
MIPI DCPHY	2; Supports DPHY or CPHY; 4-lane MIPI DPHY V2.0 (4.5 Gbps/lane); 3-lane MIPI CPHY V1.1 (2.5 Gbps/lane); Supports multiple MIPI-CSI combos
MIPI CSI DPHY	4; 2-lane MIPI DPHY V1.2 (2.5 Gbps/lane); Two 2-lane can be combined to 4-lane
DVP	1; 8/10/12/16-bit standard DVP up to 150 MHz; Supports BT.601/BT.656/BT.1120
HDMI RX	1; Supports HDMI 2.0 (3.4 ~ 6 Gbps) and HDMI 1.4b (250Mbps ~ 3.4 Gbps); HDCP2.3 and HDCP1.4
Video Output	
HDMI/eDP TX	≤2; HDMI up to 7680 × 4320@60 Hz (3/6/8/10/12 Gbps), HDCP2.3; eDP up to 4K@60 Hz (1.62/2.7/5.4 Gbps), HDCP1.3

Category/Parameter	Specification
DP TX	2; DP TX 1.4a, available for USB3.1 Gen 1, supports 1/2/4 lanes; Up to 8192 × 4320@30 Hz; HDCP2.3, HDCP1.3
MIPI DSI	2; MIPI DPHY 2.0 or CPHY 1.1, up to 4K@60 Hz; Dual MIPI-DSI left/right mode, RGB/YUV up to 10-bit
BT.1120	1; RGB up to 8-bit up to 150 MHz; Up to 1920 × 1080@60 Hz
Audio	
I2S	≤4; I2S0/I2S1: 8 lanes, TX/RX, 16-32 bits, up to 192 kHz; I2S2/I2S3: 2 lanes, TX/RX, 16~32 bits, up to 192 kHz
SPDIF	2; 2× 16-bit data storing; Biphasic stereo output
PDM	2; Up to 8 channels, 16~24 bits, up to 192 kHz; PDM primary receive mode
DSM PWM	1; Convert PCM to bitstream digital 1-bit output
Connectivity	
Ethernet	2 × GMAC by RGMII/RMII, 10/100/1000 Mbps
USB 3.1 Gen 1	3; Up to 5 Gbps; 2 × USB3.1 OTG (multiplexed with DP TX, supports USB Type-C and DP Alt); 1 × USB3.1 Host (multiplexed with PIPE PHY2)
USB 2.0 Host	2
PCIe 3.0	2; Supports RC and EP, up to 8 Gbps; Combinations: 1×4, 2×2, 4×1, 1×2+2×1
PCIe 2.0	≤3; Each supports 1 lane, up to 5 Gbps
SATA	≤3; 3 × SATA3.0 controllers, multiplexed by PCIe and USB_HOST2; Supports eSATA up to 6 Gbps

Category/Parameter	Specification
UART	≤10; 2 × 64-bit FIFO for TX/RX; 5/6/7/8-bit serial, up to 4 Mbps; All 10 UARTs support auto flow control
SPI	≤5; Each controller supports two chip select outputs; Master and slave mode configurable
I2C	≤9; 7-bit and 10-bit address modes; Standard mode up to 100 kbps, high-speed mode up to 400 kbps
PWM	≤16; Supports capturing mode
ADC	≤8; 8 × 12-bit single-end input SAR-ADC, up to 1 MS/s
Power	
Input Power	DC 4V
Mechanical	
Dimensions (W × D × H)	68 × 60 × 4.6 mm (2.68 × 2.36 × 0.18 in)
Package	Board-to-board connector (4 × 100-pin, 0.4 mm pitch, combined height 1.5 mm)
Mounting Holes	4 × φ2.5 mm
Environment	
Operating Temperature	0 °C ~ +80 °C (32 °F ~ +176 °F)

4. Software Specifications

Category/Parameter	Specification
Operating System	

Category/Parameter	Specification
OS	Linux
OS Flashing Method	USB OTG
Data Acquisition Protocol (DSA)	
Industrial Protocol	Modbus RTU Master/Slave, Modbus TCP Master/Slave, EtherNet/IP, ISO on TCP, OPC UA Client/Server, Mitsubishi MC 3C/3E/3C OverTCP, Mitsubishi CPU Port, FINS UDP, Host Link, PPI
Electricity Protocol	DLT645-2007, IEC101/104, DNP3.0
Other Protocol	BACnet, CNC
Maintenance and Management	
Upgrade Method	Supports patent upgrade mechanism, local or remote firmware upgrade
Log	Supports local system logs, remote logs, important log power-off preservation
Remote Management	InHand DeviceLive, HTTP, HTTPS, SSH, etc.
DeviceLive Cloud	Supports cloud-based parameter configuration, container management, application and firmware management

5. Ordering Information

Model Code

Model code: EC3588-C

Product Model

Model	CPU	CPU Clock Speed	Operating Temperature
EC3588-C	4 × A76 + 4 × A55	A76 @ 2.4 GHz, A55 @ 1.8 GHz	0 °C ~ +80 °C (32 °F ~ +176 °F)

6. Contact Us

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